

NextRAN-AI – 06/02/2026

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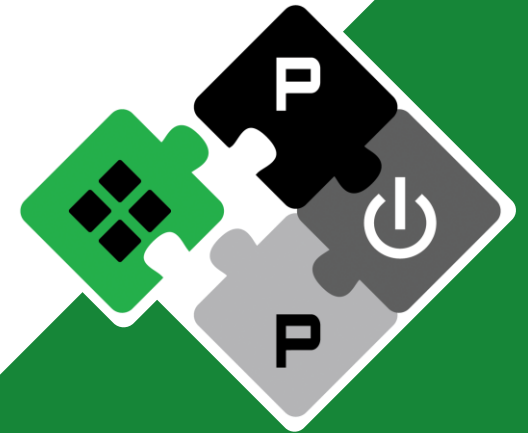
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PULP Platform

Open Source Hardware, the way it should be!



pulp-platform.org

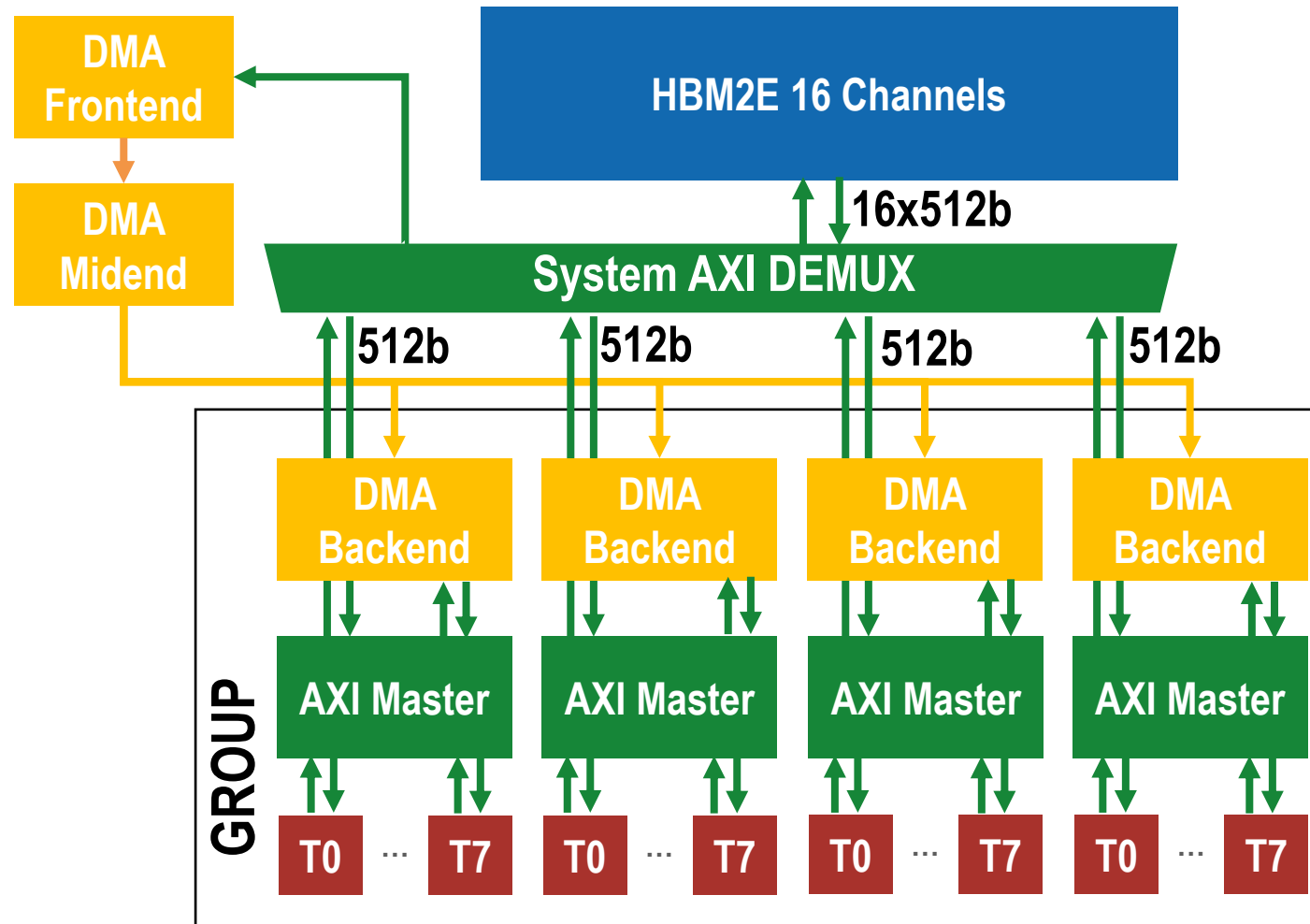
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TeraPool is connected to HBM for up to 900Gbps uplinks



- Each core can program the **DMA-Frontend** through AXI-writes
- The **Backend** initiates transfers from DRAM to each SubGroup

Memory-Compute balance w.r.t. L2



$$\pi_{TES} = 256 \frac{\text{MAC}}{\text{cycles}} \times 16 = 8192 \frac{\text{MAC}}{\text{cycles}}$$

$$\beta = 64 \frac{B}{\text{cycle}} \times N_{SG} = 1024 \frac{B}{\text{cycle}}$$

$$\frac{W}{\pi_{TES}} \geq \frac{Q_m}{\beta}$$

$$\frac{W}{\pi_{TES}} = \frac{512^3}{8192} \text{cycles} = 16384 \text{cycles}$$
$$\frac{Q_m}{\beta} = \frac{2\text{MiB}}{1024B} \text{cycles} = 2048 \text{cycles} \checkmark$$

Let's consider W (total number of MACs) and Q_m (total number of bytes) for a GEMM problem, let's assume double buffering:

$$W = M \times N \times P = M^3$$

$$Q_m = 2 \times (M \times N + N \times P + 2 \times M \times P) = 8 \times M^2 = 2\text{MiB} \rightarrow M = 512$$

Memory-Compute balance w.r.t L1 (Local)



For the Balance principle applied to local memory we consider the Tile containing one RedMulE, where memory can be accessed by the accelerator in 1-cycle:

$$\begin{aligned}\pi_{Local} &= 256 \text{ MAC/cycle} \\ \beta_{Local} &= 64 \text{ B/cycle}\end{aligned}$$

$$\begin{aligned}\text{Local Mem. Size} &= 32 \text{ Banks} \times 2\text{KiB} - 2\text{KiB}(\text{stack}) = 30\text{KiB} \\ 8 \times M^2 &= 15\text{KiB} \rightarrow M = 44\end{aligned}$$

$$\begin{aligned}T_{processing} &= 332 \text{ cycles} \\ T_{memory} &= 240 \text{ cycles} \checkmark\end{aligned}$$

Memory-Compute balance w.r.t L1 (Remote)



- Load an $L \times H \times (P+1)$ Tile in Y
- Load $L \times N$ elements from X and $H \times (P+1)$ elements from W
- Compute an $L \times H \times (P+1)$ Tile in Z

$$W = L \times H \times (P + 1) \times N = 1024N$$

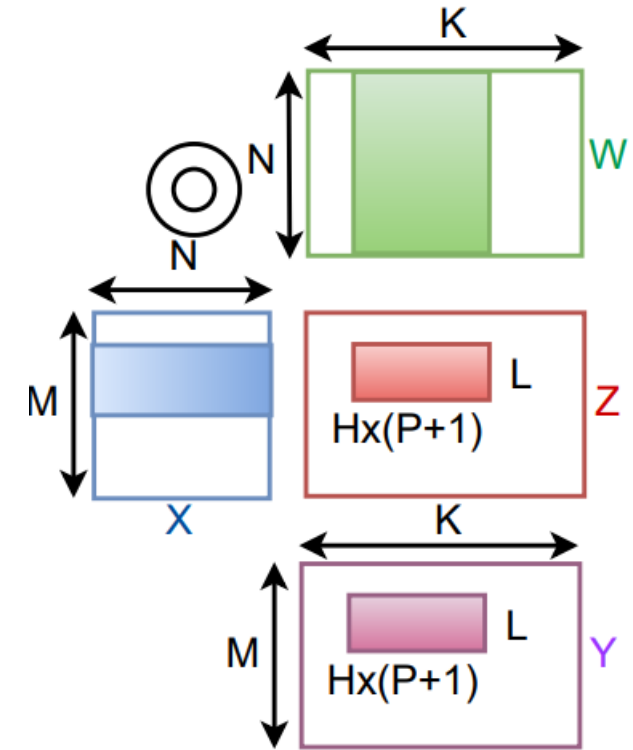
$$\pi = L \times H = 256 \text{ MAC/cycle}$$

$$\text{Store to } Z \rightarrow L \times H \times (P + 1) \times 2B$$

$$\text{Load from } Y \rightarrow L \times H \times (P + 1) \times 2B$$

$$\text{Load from } X \rightarrow L \times N \times 2B$$

$$\text{Load from } W \rightarrow H \times (P + 1) \times N \times 2B$$





Memory-Compute balance w.r.t L1 (Remote)

- Load an $L \times H \times (P+1)$ Tile in Y
- Load $L \times N$ elements from X and $H \times (P+1)$ elements from W
- Compute an $L \times H \times (P+1)$ Tile in Z

$$T_{memory} = \frac{(H \times (P + 1) + L) \times N \times 2B + L \times H \times (P + 1) \times 2B}{\beta_R} + \frac{L \times H \times (P + 1) \times 2B}{\beta_W}$$

$$T_{memory} = \frac{N \times 128B + 2048B}{\beta_R} + \frac{2048B}{\beta_W}$$

$$T_{compute} = \frac{1024N}{256} \text{ cycles}$$

$$4N \text{ cycles} > \frac{N \times 128B + 2048B}{\beta_R} + \frac{2048B}{\beta_W}$$

Memory-Compute balance w.r.t L1 (Remote)



- We have 16 local 32b ports $\rightarrow$ 64B/cycles
- We have 7 remote ports: 4 to the same SubGroup and 3 to the remote Groups
- N is large enough to assume random burst-access

$$\beta_R = 64 \frac{B}{\text{cycle}} \times \frac{32}{2048} + 4 \frac{B}{\text{cycle}} \times K \times \left(1 - \frac{32}{2048}\right) = 64 \frac{B}{\text{cycle}} \times 1.56\% + 16 \frac{B}{\text{cycle}} \times 98.4\% = 16.75$$

$$\beta_W = 64 \frac{B}{\text{cycle}} \times \frac{32}{2048} + 4 \frac{B}{\text{cycle}} \times J \times \left(1 - \frac{32}{2048}\right) = 64 \frac{B}{\text{cycle}} \times 1.56\% + 8 \frac{B}{\text{cycle}} \times 98.4\% = 8.87$$

But the burst reads can target different ports, while in this calculation only one remote port is active. The calculation does not take into account that we have outstanding requests.

Memory-Compute balance w.r.t L1 (Remote)



- With 16 outstanding burst requests of 512b each we can target 256 banks
- There are 128 banks per SubGroup:
 - **Maximum 2 read ports and minimum 1 are active at a time**
- Moreover one has to consider that we do not only fetch from one matrix but from three and each matrix has its own queue of outstanding requests:
 - **Maximum 6 read ports and minimum 1 are active at a time**

To simplify let's consider just the X and W accesses, the Y and Z can be considered negligible for large N:

$$T_{memory} = \frac{N \times 128B}{\beta_R} + \frac{2048B}{\beta_R} + \frac{2048B}{\beta_W} < 4N$$

$\beta_R > 32$

Memory-Compute balance w.r.t L1 (Remote)



What is the probability that two requests target the same port?

$$P = \frac{3}{4} \times \frac{1}{4} + \frac{1}{4} \times \left(\frac{1}{4}\right)^2 = 0.20$$

↓ 1/4 ↓ 1/4 ↓ 1/4 ↓ 1/4
↓ ↓ ↓ ↓

Not only... Since we have 16 outstanding transactions for X and 16 outstanding transactions for W it means that we have to choose 16 times the same port to not have 2 ports active...

$$\begin{aligned} \beta_R &= 64 \frac{B}{\text{cycle}} \times \frac{32}{2048} + 4 \frac{B}{\text{cycle}} \times K \times \left(1 - \frac{32}{2048}\right) \times (\mathbf{0.2^{16}} + (1 - \mathbf{0.2^{16}}) \times 2) \\ &= 64 \frac{B}{\text{cycle}} \times 1.56\% + 16 \frac{B}{\text{cycle}} \times 98.4\% \times 2 = \mathbf{33 \checkmark} \end{aligned}$$

PPA?



1. What's the profiling results regarding the utilization of TE, PE, and DMA? & the area / power figures,
2. we supposed those inputs would be the key factors to arrange the capabilities of PE/TE/DMA/MemBandwidth and parallelism properly to achieve the optimal PPA?

PPA?

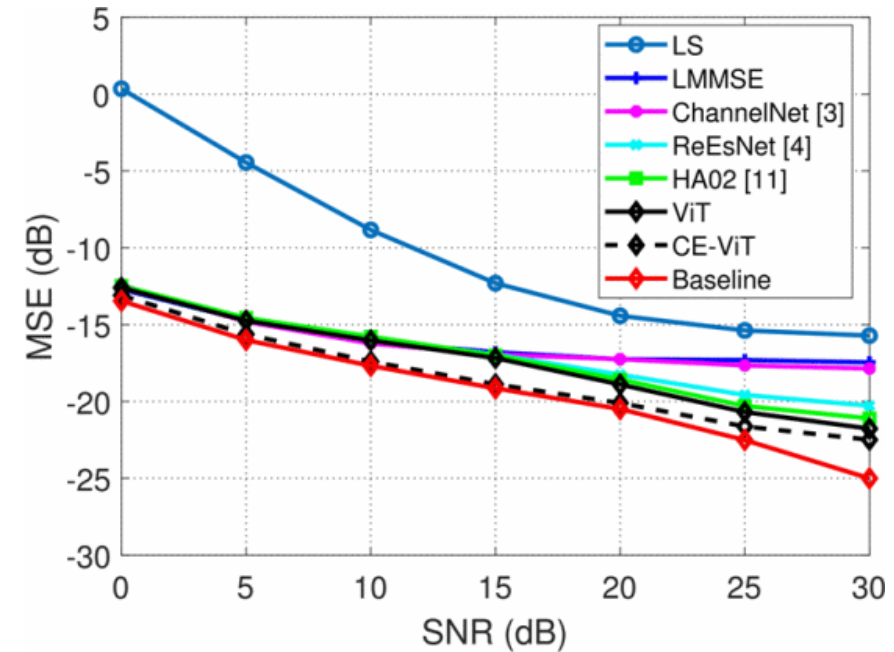


	TE	PE	Total	DMA
Peak Performance [FLOPs/cycle]	8192	1024	9216	-
Performance [FLOPs/cycle]	4,754.46	58.64	4,813.11	-
Peak Bandwidth [B/cycle]	-	-	-	256
Effective Bandwidth [B/cycle]	-	-	-	99.41
Our Focus				
Utilization Compute	58.04%	5.73%	52.23%	-
Utilization BW	-	-	-	38.83%
Instructions/cycle	-	37.84%	-	-
Area [mm2] (Cell area based on SubGroup PnR)	2.43	1.43	14.56	0.45
Power [W] (Based on Power Sim.)	4.3	1.21	5.806	0.52
Negligible				
FLOPs/cycle/mm2	1,953.15	40.87	330.57	-
FLOPs/cycle/W	1,105.69	48.47	828.99	-
FLOPs/cycle/W/mm2	454.22	33.78	56.94	-

Are we sure that this much computation is necessary?



Algorithm	Parameters	FLOPs
LS [1]	-	0.044M
LMMSE [1]	-	136.78M
ChannelNet [3]	0.68M	2299.89M
ReEsNet [4]	0.18M	71.18M
HA02 [11]	0.22M	23.92M
ViT	0.14M	24.13M
CE-ViT	0.23M	24.42M

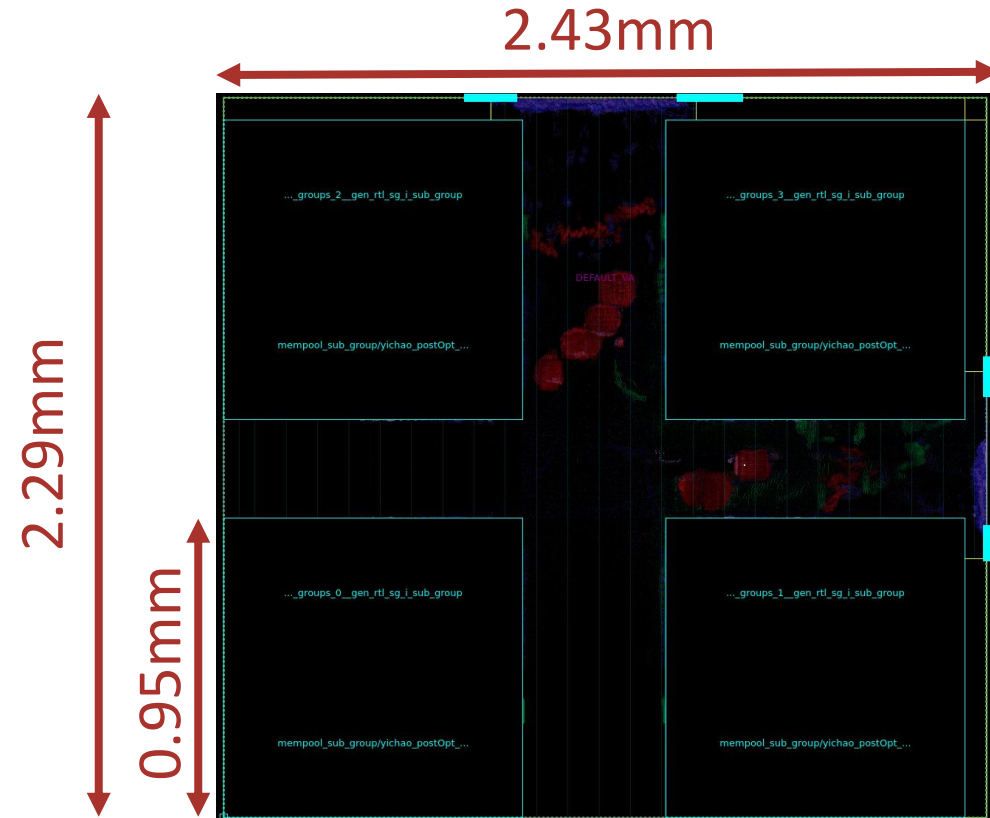


«CE-ViT: A Robust Channel Estimator Based on Vision Transformer for OFDM Systems»

(<https://ieeexplore.ieee.org/document/10436847>)

Group: PnR TSMC-N7

- 8x32 RedMulE, 16-ROB entries, REQ-GRP = 2, RSP-GRP = 4
- 4 Tiles / SubGroup, 4 SubGroups / Group, 4 Groups / Cluster
- Area SubGroup = 0.9 mm², Area Group = 5.6 mm², Area channels = 2 mm² → **55% loss in area efficiency**



Group: PnR TSMC-N7



	TensorPool (N7)	TeraPool (N7)*	Improvement
Area** [mm ²]	22.4	27.8	
Area SubGroup [mm ²]	0.9	1.0	
Area Group [mm ²]	5.6 (+55%)	5.9 (+47%)	
T [ns]	1.1	1.1	
Peak FLOPs/cycle (TEs)	8,192	0	
Peak FLOPs/cycle (TEs + PEs)	9,216	4,096	2.25x
Peak GFLOPs/s	8,378.18	3723	1.54x
Peak GFLOPs/s/mm ²	374	134	2.79x

* Technology normalization from GF12, ** Based on the Group area

Pin alignment on 3D integration

- Upon flipping the Group macro the only option to have pin alignment is to create two separate macros
- Group macros are treated as **black-boxes**

